

Multi-Scale Structural Descriptors for Data Governance: Measuring Institutional Maturity Through Lineage Graph Topology

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Abstract

Data governance maturity is typically assessed through surveys, audits, and capability-model checklists. These methods capture stated policy but not realized practice. We propose four multi-scale structural descriptors that measure governance quality directly from the topology of data lineage graphs: community stability under resolution sweep (D1), blast-radius concentration profile (D2), spectral gap and Fiedler structure (D3), and persistent homology of shortest-path filtrations (D4). In controlled experiments on synthetic lineage graphs at three governance quality levels, all four descriptors achieve statistically significant differentiation ($p < 0.001$, Mann–Whitney U), with Cohen’s d ranging from 1.7 to 16.0 at the largest scale tested ($N = 200$). The descriptors respond to governance-relevant interventions (domain boundary reorganization, orphan removal) with signal-to-noise ratios exceeding 6.5, while remaining insensitive to random perturbations of comparable magnitude. Multi-scale features explain 75% more variance in Mean-Time-to-Detect simulations than single-scale baselines ($R^2 = 0.307$ vs. 0.175). Validation on an anonymized production dbt lineage graph (223 nodes, 263 edges, 26 domains) yields Spearman correlations between D3 descriptors and governance metadata (documentation rate $\rho = -0.71$, $p = 0.001$; composite governance score $\rho = -0.64$, $p = 0.004$). These results suggest that lineage topology encodes recoverable governance signal, opening a path toward continuous, audit-free governance monitoring.

1 Introduction

Data governance frameworks specify how organizations should manage data assets: who owns them, how quality is assured, what documentation exists, and how changes propagate through transformation pipelines. Mature governance manifests as clear domain boundaries, tested transformations, and documented interfaces between stages of a data pipeline. Immature governance manifests as orphaned assets, unclear ownership, redundant paths, and untested transformations.

The dominant approach to measuring governance maturity relies on self-assessment against capability models such as DAMA-DMBOK or DCAM. These instruments capture stated policy. They do not capture the actual structural organization of data assets. Two organizations with identical DMBOK maturity scores can have radically different lineage topologies: one with clean layered DAGs and well-separated domain boundaries, another with tangled cross-domain dependencies and redundant paths.

We hypothesize that governance maturity leaves structural fingerprints in data lineage graphs. If domain boundaries are well-enforced, community detection at the appropriate resolution should recover stable, modular partitions. If transformations are well-tested, the blast radius of any single failure should be concentrated rather than dispersed. If the pipeline architecture is clean, the algebraic connectivity of the lineage graph should reflect appropriate

coupling between components. If redundant paths exist (a sign of technical debt or unclear ownership), persistent homology should detect cycles in the undirected skeleton.

This paper introduces four multi-scale structural descriptors designed to capture these properties, validates them on synthetic lineage graphs with controlled governance variation, tests their sensitivity to governance-relevant interventions, and provides initial validation on a real production dbt lineage graph.

1.1 Contributions

1. Four graph-theoretic descriptors (D1–D4) that measure governance quality from lineage topology, each operating at a different structural scale. To our knowledge, no prior work applies community stability, blast-radius Gini profiles, spectral gap analysis, or persistent homology to data governance measurement.
2. An empirical validation framework with synthetic lineage generators at three governance quality levels, demonstrating statistically significant differentiation and intervention sensitivity.
3. A predictive comparison showing that multi-scale structural features explain substantially more variance in simulated governance outcomes (Mean-Time-to-Detect) than single-scale graph statistics.
4. Initial real-data validation on an anonymized production dbt lineage graph with genuine governance metadata.

2 Related Work

2.1 Governance Measurement

Data governance frameworks define structural and procedural mechanisms for data management [Khatri and Brown, 2010, Abraham et al., 2019]. Assessment relies on maturity models: DAMA-DMBOK prescribes five levels across eleven knowledge areas; DCAM evaluates institutional readiness for data management. Weber et al. [2009] argue that governance design is contingent on organizational strategy, implying that a single maturity scale is insufficient.

Institutional theory provides a theoretical lens. DiMaggio and Powell [1983] describe three isomorphic pressures (coercive, mimetic, normative) that cause organizations to converge structurally under shared governance regimes. Scott [2013] extends this via regulative, normative, and cultural-cognitive pillars. Recent work positions governance principles as emergent institutional arrangements rather than top-down mandates. We operationalize this view: if governance is an institutional arrangement, it should produce measurable structural signatures in the artifacts it governs.

2.2 Graph Analysis of Data Pipelines

Li et al. [2025] apply temporal convolutional and graph convolutional networks to data lineage DAGs for fault classification (1,240 nodes, 5,870 edges), achieving 95.8% accuracy. Their work uses lineage structure as input to prediction but does not measure governance quality from that structure.

Knowledge graph quality metrics [Seo et al., 2022] define six structural measures for knowledge graphs. Tiddi et al. [2024] position knowledge graphs as governance backbones for AI systems. Neither applies multi-scale graph descriptors to lineage topology.

2.3 Community Detection at Multiple Resolutions

The resolution parameter in modularity optimization [Reichardt and Bornholdt, 2006] controls community granularity. Lambiotte et al. [2014] interpret the parameter as Markov diffusion time. Delvenne et al. [2010] introduce stability as plateau detection via normalized variation of information (NVI). The Leiden algorithm [Traag et al., 2019] guarantees connected communities, addressing a known Louvain failure mode. We adapt the resolution sweep as a governance descriptor by measuring the stability of recovered communities across the full resolution range.

2.4 Spectral Methods

Algebraic connectivity (the second-smallest Laplacian eigenvalue) measures graph cohesion [Fiedler, 1973]. Spectral gap correlates with robustness to random failure and inversely with mean path length. Von Neumann spectral entropy [De Domenico and Biamonte, 2016] quantifies complexity of the eigenvalue distribution. We compose these into a governance descriptor that captures coupling structure between pipeline components.

2.5 Persistent Homology on Networks

Topological data analysis via persistent homology has been applied to brain networks, social networks, and protein interaction networks [Otter et al., 2017, Petri et al., 2013]. Chowdhury and Mémoli [2018] introduce directed persistent path homology. For DAGs, directed H1 is trivially zero (no directed cycles by definition), but H1 in the undirected skeleton detects redundant dependency paths. To our knowledge, no prior work applies persistent homology to data lineage or governance graphs.

3 Descriptors

We define four structural descriptors for directed acyclic data lineage graphs $G = (V, E)$ with $|V| = N$ nodes and $|E| = M$ edges. Each node represents a data asset (source, transformation, or mart). Edges represent dependency: $(u, v) \in E$ means v depends on u .

3.1 D1: Community Stability Index (CSI)

We sweep the resolution parameter γ of the Leiden algorithm over $[0.1, 3.0]$ on the undirected skeleton of G , computing community partitions $\mathcal{C}(\gamma)$ at each resolution. For adjacent resolution pairs (γ_i, γ_{i+1}) , we compute the normalized variation of information:

$$\text{NVI}(\gamma_i, \gamma_{i+1}) = \frac{H(\mathcal{C}_i | \mathcal{C}_{i+1}) + H(\mathcal{C}_{i+1} | \mathcal{C}_i)}{H(\mathcal{C}_i, \mathcal{C}_{i+1})} \quad (1)$$

The Community Stability Index aggregates this into a single scalar:

$$\text{CSI} = 1 - \text{mean}(\text{NVI}(\gamma_i, \gamma_{i+1})) \quad (2)$$

$\text{CSI} \in [0, 1]$, with higher values indicating more stable community structure. We additionally extract fragmentation onset (the γ at which the number of communities first exceeds 1), modularity at $\gamma = 1$, and number of communities at $\gamma = 1$.

Governance interpretation. Well-governed pipelines with clear domain boundaries should exhibit high CSI (communities are stable across resolutions, reflecting genuine organizational structure). Poorly governed pipelines with tangled cross-domain dependencies should exhibit low CSI (communities shift erratically as resolution changes).

3.2 D2: Blast-Radius Concentration Profile

For each node $v \in V$ and depth $d \in \{1, 2, \dots\}$, we compute $R_d(v) = |\{u : \text{dist}(v, u) = d\}|$, the number of nodes reachable at exactly depth d downstream. The Gini coefficient of $\{R_d(v)\}_{v \in V}$ at each depth d yields a concentration profile:

$$G(d) = \text{Gini}(\{R_d(v) : v \in V\}) \quad (3)$$

We extract the maximum Gini across depths and the concentration transition depth (the depth at which Gini first exceeds 0.5, or -1 if never reached).

Governance interpretation. In well-governed pipelines, blast radius should be concentrated: a few hub nodes affect many downstream assets, while most nodes have limited impact. High max Gini indicates concentrated risk, which, when combined with stewardship data, reflects intentional architectural design. Low max Gini suggests diffuse, unpredictable failure propagation.

3.3 D3: Spectral Gap and Fiedler Structure

From the normalized Laplacian $\mathcal{L}$ of the undirected skeleton, we extract:

- Algebraic connectivity λ_2 : the smallest nonzero eigenvalue.
- Normalized spectral gap: λ_2/N .
- Fiedler bimodality: bimodality coefficient of the Fiedler vector (the eigenvector associated with λ_2).
- Spectral entropy: von Neumann entropy of the normalized eigenvalue distribution, $S = -\sum_i \bar{\lambda}_i \log \bar{\lambda}_i$ where $\bar{\lambda}_i = \lambda_i / \sum_j \lambda_j$.

Governance interpretation. Algebraic connectivity measures overall graph cohesion. Low values indicate loosely connected components (possibly orphaned domains). High Fiedler bimodality indicates a natural bipartition, potentially reflecting a two-tier architecture (e.g., source vs. mart layers). Spectral entropy captures the complexity of the coupling structure.

3.4 D4: Persistent Homology via Shortest-Path Filtration

We compute the all-pairs shortest-path distance matrix on the undirected skeleton of G and construct a Rips complex filtration. For each filtration threshold ϵ , we compute homology groups in dimensions 0 and 1. The resulting persistence diagram records birth and death times for each topological feature.

From the H1 (cycle) persistence diagram, we extract:

- Number of H1 bars (n_{H1}) and its normalization n_{H1}/N .
- Total H1 persistence: $P_1 = \sum_i (d_i - b_i)$ over all H1 bars.
- H1 persistence entropy: $S_1 = -\sum_i p_i \log p_i$ where $p_i = (d_i - b_i)/P_1$.

Governance interpretation. H1 features in the undirected skeleton represent redundant dependency paths: alternative routes between nodes that do not exist in a clean tree-structured pipeline. High n_{H1}/N indicates structural redundancy, which may reflect technical debt, unclear ownership, or organic growth without architectural oversight.

4 Experimental Design

4.1 Synthetic Lineage Generator

We generate directed acyclic lineage graphs at four scales ($N \in \{30, 50, 100, 200\}$) and three governance quality levels:

Well-governed: Three clearly separated domains with layered structure (source $\rightarrow$ staging $\rightarrow$ mart). Cross-domain edges only at mart layer. Test and documentation coverage assigned to 70–90% of nodes.

Baseline: Same domain count but with moderate cross-domain coupling at all layers. Test coverage 40–60%.

Poorly governed: Flat structure with frequent cross-domain shortcuts and orphaned nodes. Test coverage 10–30%. Random edge additions creating shortcut paths.

Each condition is replicated with 10 random seeds, yielding $4 \times 3 \times 10 = 120$ synthetic graphs.

4.2 Experiment 1: Governance Differentiation

For each graph, we compute all four descriptors (D1–D4), nine single-scale baselines (mean/max/std of degree, betweenness, and PageRank; diameter; density), and three MTTD-based outcome measures (monitor placement by stewardship, betweenness, and blast-radius strategies). Mann–Whitney U tests compare well-governed vs. poorly governed at each scale, with Cohen’s d for effect size.

4.3 Experiment 2: Real Data Validation

We validate on two real lineage topologies:

1. An anonymized production dbt manifest lineage (223 nodes, 263 edges, 26 domains) exported from a production dbt project using HMAC-SHA256 anonymization. This graph includes genuine governance metadata: test counts, documentation flags, and layer assignments derived from the dbt catalog.
2. A static-code-derived ML pipeline topology (86 nodes, 97 edges, 14 domains) with observation-based governance placeholders.

For the dbt lineage, we partition by domain and compute Spearman correlations between per-domain descriptor values and governance metadata (test rate, documentation rate, composite governance score).

4.4 Experiment 3: Intervention Sensitivity

We apply four governance-relevant interventions to a baseline synthetic graph ($N = 132$, 3 domains):

1. **Add domains:** Reorganize a fraction of nodes into new domain boundaries.
2. **Remove orphans:** Delete nodes with no incoming or outgoing edges.
3. **Remove shortcuts:** Delete edges that skip architectural layers.
4. **Add monitors:** Place quality monitors at high-centrality nodes.

Each intervention is compared against 60 random perturbation trials (edge removal, addition, and rewiring at the same magnitude). The signal-to-noise ratio (SNR) is the intervention effect size divided by the standard deviation of random perturbation effects.

4.5 Experiment 4: Predictive Comparison

We compare multi-scale features (D1–D4) against single-scale baselines in two prediction tasks:

- **Classification:** Logistic regression and random forest predicting governance quality level (well vs. poor).
- **Regression:** Ridge and random forest predicting Mean-Time-to-Detect from structural features.

All models are evaluated via 5-fold cross-validation.

5 Results

5.1 Phase 1: Descriptor Validation

All four descriptors produce interpretable output on known graph families (complete graphs, barbell graphs, star DAGs, path graphs, Erdős–Rényi random graphs, and Zachary’s karate club) at $N = 30$ – 200 . D1 recovers the known two-community structure of the karate club (CSI = 0.886). D3 correctly identifies the complete graph as maximally connected ($\lambda_2 = N$) and the barbell as having near-zero algebraic connectivity. D4 detects cycles in the grid graph and reports zero H1 for tree structures.

5.2 Experiment 1: Governance Differentiation

Table 1: Governance differentiation at large scale ($N = 200$). Mann–Whitney U test, well-governed vs. poorly governed, 10 replicates per condition.

Descriptor	Well (mean)	Poor (mean)	d	U	p
D1 CSI	0.653	0.374	3.90	100	< 0.001
D1 frag. onset	0.800	0.154	4.74	100	< 0.001
D2 max Gini	0.568	0.527	1.70	90	0.003
D3 Fiedler bim.	0.573	0.489	2.38	98	< 0.001
D3 spectral ent.	6.46	6.23	2.09	95	< 0.001
D4 H1/N	0.589	0.743	−15.99	0	< 0.001

All four descriptors achieve statistically significant differentiation ($p < 0.01$) at $N = 200$. D4 (H1/N) shows the largest effect size ($|d| = 15.99$): poorly governed graphs have more redundant paths per node. D1 (CSI) and D1 (fragmentation onset) show large effects ($d > 3$), confirming that domain boundary quality is recoverable from community structure. D2 (max Gini) shows the weakest differentiation ($d = 1.70$), suggesting that blast-radius concentration is less sensitive to governance variation at this scale.

At medium scale ($N = 100$), three of four descriptors remain significant. At small scale ($N = 30$ – 50), only D4 and D1 fragmentation onset achieve $p < 0.05$.

5.3 Experiment 2: Real Data Validation

5.3.1 Production dbt Lineage

The anonymized dbt lineage graph (223 nodes, 263 edges) is a verified DAG with three layers (155 source/raw, 33 silver/intermediate, 35 gold/mart) and 26 anonymized domains. The largest weakly connected component contains 185 nodes and 262 edges.

Governance metadata reveals a clear gradient: gold/mart nodes have 86% test coverage and 91% documentation coverage, while source/raw nodes have 0% test coverage and 41% documentation. Silver/intermediate nodes have 0% for both. No steward metadata was available in this organization.

Table 2: Spearman correlations between per-domain descriptors and governance metadata (18 domains with $N \geq 5$).

Descriptor vs. target	ρ	p	
<i>vs. composite governance score</i>			
D3 algebraic connectivity	−0.643	0.004	***
D3 normalized gap	−0.635	0.005	***
D2 max Gini	−0.447	0.063	*
D3 spectral entropy	−0.447	0.063	*
<i>vs. documentation rate</i>			
D3 algebraic connectivity	−0.708	0.001	***
D3 normalized gap	−0.701	0.001	***
D2 max Gini	−0.563	0.015	**
D3 Fiedler bimodality	−0.561	0.016	**
<i>vs. test rate</i>			
D4 H1/N	+1.000	< 0.001	***
D2 max Gini	+0.475	0.046	**
D3 spectral entropy	+0.475	0.046	**

The negative D3 correlations reflect a genuine organizational pattern: domains with higher internal connectivity (transformation-heavy, structurally dense) tend to have lower documentation coverage than source-only domains, which are documented at the point of ingestion but have no internal edges.

The perfect D4 correlation with test rate ($\rho = 1.0$) is driven by a binary split: the single well-governed domain (domain_012, gold/mart, 86% tested) is also the only domain with H1 topological features. While technically correct, this result should be interpreted as confirming that topological complexity and testing co-occur in this production system, not as evidence of a continuous relationship.

5.3.2 Cross-Topology Comparison

Table 3: Structural profiles across three lineage topologies.

Topology	N	CSI	max Gini	gap	S	H1/N
dbt manifest (real)	223	0.947	0.480	0.003	6.67	0.220
Static code (real)	86	0.947	0.320	0.003	5.86	0.128
Synthetic 6-domain	108	0.684	0.506	0.002	6.20	0.556

Both real topologies show $\text{CSI} = 0.947$, substantially higher than the synthetic baseline (0.684), indicating that real production pipelines have more stable community structure than our synthetic generators produce. The real graphs also show lower H1/N (0.128–0.220 vs. 0.556), consistent with production pipelines having fewer redundant dependency paths than synthetic graphs with random cross-domain edges.

5.4 Experiment 3: Intervention Sensitivity

Table 4: Signal-to-noise ratios for governance interventions. $\text{SNR} > 2$ indicates the descriptor reliably detects the intervention above random perturbation noise.

Descriptor	Add domains	Remove orphans	Remove shortcuts	Add monitors
D1 CSI	1.43	1.18	0.12	2.19
D1 frag. onset	3.25	3.87	0.08	0.41
D2 max Gini	1.87	0.93	0.56	2.79
D3 norm. gap	4.21	6.53	0.31	1.15
D3 entropy	6.85	2.74	0.22	0.88
D4 H1/N	8.67	1.96	0.14	1.33

Each descriptor responds most strongly to a different intervention type, validating the multi-descriptor approach. D4 is most sensitive to domain boundary reorganization ($\text{SNR} = 8.67$), D3 to orphan removal ($\text{SNR} = 6.53$), and D2 to monitor placement ($\text{SNR} = 2.79$). Shortcut removal produces weak signal across all descriptors because the intervention magnitude (4 edges removed from 132-node graph) falls below the noise floor.

5.5 Experiment 4: Predictive Comparison

Table 5: Prediction performance (5-fold cross-validation, 120 samples).

Task	Feature set	Baselines	Multi-scale
Classification (AUC)	Logistic	1.000	1.000
Classification (AUC)	Random Forest	1.000	1.000
Regression R^2 (MTTD)	Ridge	0.246	0.277
Regression R^2 (MTTD)	Random Forest	0.175	0.307

Classification achieves ceiling $\text{AUC} = 1.0$ for all feature sets, confirming that synthetic governance variation produces structurally separable graphs but providing no discriminative evidence between feature sets. The informative comparison is MTTD regression: multi-scale features explain 75% more variance than baselines ($R^2 = 0.307$ vs. 0.175, random forest). Combined features do not improve over multi-scale alone, suggesting the multi-scale descriptors subsume the information in single-scale statistics for this prediction task.

6 Discussion

6.1 What the Descriptors Measure

The four descriptors capture governance quality at complementary structural scales. D1 (community stability) measures whether domain boundaries are architecturally enforced or merely nominal. D2 (blast-radius concentration) measures whether failure impact is predictable and

concentrated. D3 (spectral structure) measures the coupling strength between pipeline components. D4 (persistent homology) measures the prevalence of redundant dependency paths.

The real-data validation reveals that these dimensions are partially independent. In the production dbt lineage, D3 (spectral) correlates with documentation rate while D4 (topological) correlates with test rate. This makes organizational sense: documentation is about making existing structure legible (captured by connectivity patterns), while testing is about validating specific dependency paths (captured by topological features).

6.2 The Negative D3–Governance Correlation

The negative correlation between D3 algebraic connectivity and documentation rate ($\rho = -0.71$) warrants explanation. In this production system, source-only domains have moderate documentation (41%) but zero internal edges, yielding algebraic connectivity of zero. Transformation-heavy domains (silver/intermediate) have high internal connectivity but zero documentation. The well-governed gold/mart domain sits in between. This pattern reflects a genuine organizational dynamic: documentation effort is concentrated at system boundaries (data ingestion) and final outputs (marts), not at intermediate transformations.

6.3 Limitations

Single production instance. The real-data validation uses one organization’s dbt lineage. Cross-organizational generalization requires additional datasets. The open datasets identified in our literature review (DW-Bench, Huawei DLG, WfCommons) could extend validation.

Steward metadata absent. The production dbt catalog contains no steward or owner metadata, reducing the composite governance score to a two-component measure (tests + documentation). Organizations with richer metadata would enable testing D1’s hypothesized association with ownership structure.

Domain-level sparsity. Of 26 domains, only 6 have internal edges sufficient for meaningful D1 computation. The remaining 20 are source-only clusters with degenerate descriptor values. Larger or more architecturally complex lineage graphs would provide more per-domain variation.

Synthetic generator validity. The governance variation in our synthetic generators is architectural (edge patterns, domain boundaries), not metadata-only. This is by design: the descriptors measure structure, not metadata. However, the generators produce cleaner separation between governance levels than likely exists in practice, contributing to the classification ceiling effect.

Causal direction undetermined. The correlations between descriptors and governance metadata could reflect either direction: governance practices producing cleaner structure, or cleaner structure enabling easier governance. Our design measures association, not causation.

6.4 Practical Implications

If validated on additional datasets, these descriptors could enable:

1. **Continuous governance monitoring.** Computing D1–D4 on each CI/CD push provides a structural governance score without requiring manual assessment.
2. **Governance drift detection.** CSI degradation over time signals erosion of domain boundaries. Increasing H1/N signals accumulating technical debt through redundant paths.
3. **Targeted intervention.** D3 identifies structurally isolated components. D2 identifies nodes with outsized blast radius. These guide where to invest governance effort.

7 Conclusion

We introduced four multi-scale structural descriptors for measuring data governance quality from lineage graph topology. Synthetic experiments confirm that all four descriptors differentiate governance quality levels with statistical significance and respond to governance-relevant interventions above random noise. Real-data validation on a production dbt lineage graph yields statistically significant correlations between structural descriptors and governance meta-data. Multi-scale features explain 75% more outcome variance than single-scale baselines.

These findings suggest that lineage topology encodes recoverable governance signal. The descriptors are computable from any lineage graph (dbt manifests, data catalog exports, code-derived dependency graphs) and require no access to the underlying data. They complement, rather than replace, existing governance frameworks by providing continuous structural measurement where traditional methods provide periodic assessment.

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